Course: M.Sc. (IT) – Semester II

Subject: Artificial Intelligence & Machine Learning

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Chapter 5

Knowledge Representation Structures

Introduction

Most common knowledge representation schemes fall into the following categories:

1. Structured Knowledge Representation
2. Logic based knowledge Representation
3. Procedural based knowledge Representation
4. Graph/Network knowledge Representation

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Structured Knowledge Representation

Structured representation organizes knowledge in a **well-defined format**, like tables or records.

Examples: Frames , Database (DBMS)

Frames

- A **Frame** is like a record or form that stores information about an object or concept.
- It contains:
  - ✓ Attributes (called slots)
  - ✓ Values

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Structured Knowledge Representation

Frame: **Person**

Frames group all related information in one place.

Attribute	Value
Profession	Engineer
Age	25
City	Ahmedabad
State	Gujarat

Structured Knowledge Representation

Relational Database

A relational database stores data in the form of **tables**.

Each record has multiple fields.

Example: Disease Record

Field	Value
Disease Name	Iron Deficiency
Organ System	Blood
Gender Affected	Female
Diagnostic Test	CBC

Logic Based Knowledge Representation

- Logic-based knowledge representation means **storing knowledge as logical statements that are either TRUE or FALSE**.
- Computers use this method to **reason and answer questions logically**.

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Logic Based Knowledge Representation

Step 1: Propositional Logic (Simple True/False Statements)

- In propositional logic, we represent facts using symbols like **P, Q, R**.

Example:

Let:

- **P** = MCV is decreased in iron deficiency anemia
- **Q** = MCV is increased in pernicious anemia

Here, each statement is either **true or false**.

We can combine them using logical operators:

- ✓ **P AND Q** → Both statements are true
- ✓ **P OR Q** → At least one is true
- ✓ **P AND (NOT Q)** → P is true, Q is false

- This method is simple, but it cannot describe detailed relationships.

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Logic Based Knowledge Representation

What is MCV?

MCV (Mean Corpuscular Volume) is a blood test value.

It helps doctors classify anemia into:

Microcytic anemia (low MCV)

Normocytic anemia (normal MCV)

Macrocytic anemia (high MCV)

So MCV helps in medical diagnosis.

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Logic Based Knowledge Representation

Step 2: Predicate Logic (More Powerful)

- Propositional logic is limited.
  So we use **Predicate Logic** (also called Predicate Calculus).

Instead of writing:

- P = MCV is decreased in iron deficiency anemia
- We write it in a structured form:
- MCV(decreased, iron deficiency)

Similarly:

- MCV(increased, pernicious anemia)

Here:

MCV → Predicate (relation)

decreased / increased → Property

iron deficiency / pernicious anemia → Disease

This format clearly shows the relationship between MCV value and disease.

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Logic Based Knowledge Representation

Asking Questions Using Predicate Logic

- Now the system can answer queries.

Example question: MCV(x, iron deficiency)

Here:

- x is a variable
- We are asking: "What is the MCV value in iron deficiency?"
- The knowledge base will return:
  - ✓ MCV(decreased, iron deficiency)
- So the answer is:
  MCV is decreased in iron deficiency anemia.

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Procedural Based Knowledge Representation

- Procedural knowledge representation means **knowledge is stored in the form of steps or rules**.
- It tells the system:
  - ✓ **What to do**
  - ✓ **When to do it**
- It is like giving instructions to a computer.

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Rule-Based System

- The most common procedural representation is a **Rule-Based System**.
- Rules are written in this format:
  - ✓ IF condition
  - ✓ THEN conclusion

Procedural Based Knowledge Representation

Example (Medical Diagnosis)

- IF MCV is increased
  THEN conclude pernicious anemia
- IF MCV is decreased
  THEN conclude iron deficiency anemia

Here:

- Condition → MCV value
- Action → Diagnosis

So the system follows a **process**:

1. Check MCV value
2. Apply rule
3. Give result

It works like a decision-making procedure.

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Graph / Network Knowledge Representation

In this method, knowledge is shown as a **diagram or network**.

It uses:

Nodes → Objects or concepts

Arcs (Links) → Relationship between objects

Types of Network Structures

- Frame Networks
- Semantic Networks
- Causal Networks
- Decision Trees
- Artificial Neural Networks

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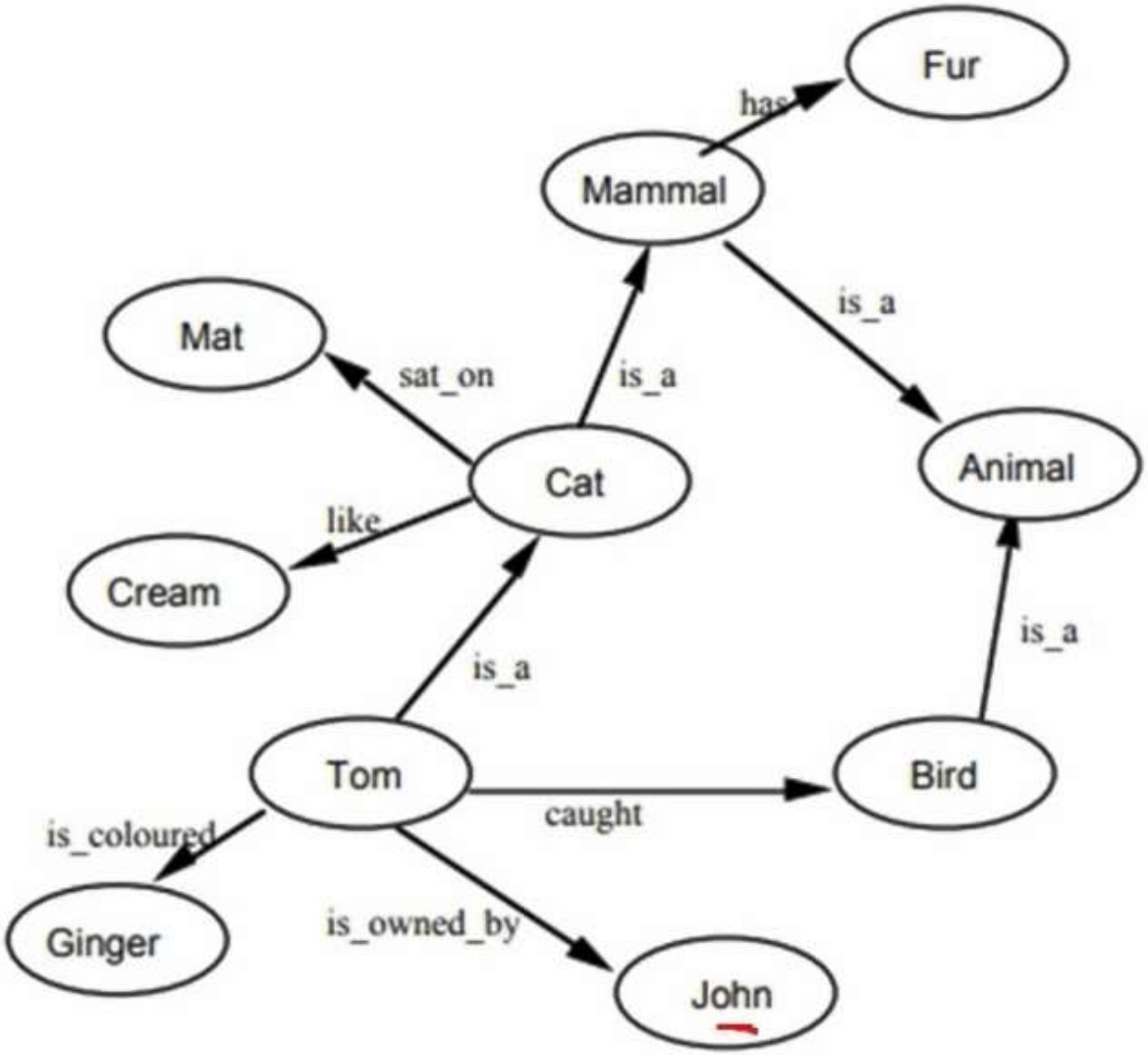
Graph / Network Knowledge Representation



In image:

- **Tom** is a **Cat**
- **Cat** is a **Mammal**
- **Mammal** is an **Animal**
- **Bird** is also an **Animal**

Relationships shown:

- Tom is_coloured Ginger
- Tom is_owned_by John
- Tom caught Bird
- Cat likes Cream
- Cat sat_on Mat
- Mammal has Fur

First Order Logic

Logical Representation can be done in two ways:

1. Propositional Logic
2. First order Predicate Logic

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Propositional Logic

- **Propositional Logic** is the simplest form of logic.
- A **proposition** is a statement that is either **True** or **False**, but not both.

Examples of Propositions

"This class is CS122." → (True)

"Today is Sunday." → (False)

Each statement must clearly have only one truth value.

Using Symbols

Instead of writing full sentences, we use symbols like **A, B, C** to represent statements.

Example:

A – It is hot

B – It is humid

C – It is raining

NOT	¬
AND	∧
OR	∨
IMPLICATION	→
IFF	⇔

Now we can write logical relationships using symbols.

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Propositional Logic

Logical Statements

- If it is humid, then it is hot:

  $B \rightarrow A$

  ($\rightarrow$ means "if…then")

If it is hot and humid, then it is not raining:

$(A \wedge B) \rightarrow \neg C$

Where:

- $\wedge$ means AND
- $\neg$ means NOT

So this statement means:

If both A and B are true, then C must be false.

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Propositional Logic

Limitation of Propositional Logic

- Propositional logic can only handle simple true/false statements.
- It **cannot represent general statements** like:
- All triangles have three sides.
- Some people are lazy.
- Because it does not use variables or quantifiers (like “all” or “some”).

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First Order Predicate Logic

In **First Order Predicate Logic (FOPL)**, every sentence is made up of two main parts:

- **Subject** → The naming part (who or what the sentence is about)
- **Predicate** → The telling part (what is happening

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First Order Predicate Logic

- **Examples of Subject and Predicate**
- I **sing**.
- He **was cooking dinner**.
- We **saw the cat outside**.
- I **walked the dog**.
- Anthony **wrote to his friend**.
- They **ate all the candy**.
- My aunt **moved**.
- The house **has a new roof**.
- Andrew **threw the ball**.
- He **is sad**.

In all these examples:

- The subject is the person/thing.
- The predicate tells the action or state.

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First Order Predicate Logic

Syntax and Semantics

- Every sentence in logic has:
- **Syntax** → Structure or form (how it is written)
- **Semantics** → Meaning (what it represents)
- To check correct structure, we use something called a **Well Formed Formula (WFF)**.

Components of First Order Logic

- An expression in FOPL contains:
- Predicate symbols
- Objects
- Variables
- Function symbols
- Constant symbols

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First Order Predicate Logic

Examples

Objects:

People, houses, numbers, games

Relations:

brother-of, bigger-than, inside, part-of

Properties:

round, prime, multi-storied

Functions:

father-of, mother-of, best-friend

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First Order Predicate Logic

Primitives Defined by User

1. Constant Symbols

- Represent specific individual objects.

Examples: John, 3

2. Function Symbols

- Show relationship between individuals.

Examples: Father-of(Reji) = John , color-of(moon) = white

3. Predicate Symbols

- Show whether something is true or false.

  Examples: greater(5,3) , white(moon)
- A predicate shows a relation between objects.

  Example: “John likes football”

  Written as: like(John, football)

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First Order Predicate Logic

Connectives and Quantifiers

To form proper logical statements (WFF), we use:

- Logical connectives (AND, OR, NOT, →)
- Quantifiers

There are two main quantifiers:

1. Universal (∀)
2. Existential (∃)

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First Order Predicate Logic

1. Universal Quantifier (∀)

- It means "for all".
- (∀x)P(x) Means P is true for every value of x.

- Example: ∀x: penguin(x) → bird(x)
  Meaning: All penguins are birds.
- Another example:
  ∀x: computer-science-student(x) → smart(x)
  Meaning: All computer science students are smart.
- Rare case:
  - ∀x: computer-science-student(x) ∧ smart(x)
    Meaning: Everyone in the world is a CS student and smart (which is unrealistic).

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First Order Predicate Logic

2. Existential Quantifier (∃)

- It means “there exists” or “at least one”.
- (∃x)P(x)
  Means P is true for some value of x.
- Example:
  ∃x: bird(x) ∧ lives-Antarctica(x)
  Meaning: There is a bird that lives in Antarctica.
- Another example:
  ∃x: computer-science-student(x) ∧ smart(x)
  Meaning: There is at least one smart CS student.

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First Order Predicate Logic

Common Mistake

Writing:

∃x: computer-science-student(x) → smart(x)

This is incorrect because it does not clearly mean "there exists a smart CS student."

Terms and Atoms

- A **Term** represents a real-world object.
  It can be:
- Constant (John)
- Variable (x)

An **Atom** is a predicate with terms.

It can be true or false.

- Example: likes(John, football)

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Connectives and Symbols

Connective	Symbols	Read As
Assertion	p	“p is true ”
Negation	¬p	“p is false”
Conjunction	p ∧ q	“both p and q are true”
Disjunction	P ∨ q	“either p is true, or q is true, or both”
Implication	P → q	“if p is true then q is true”, ”p implies q”
Equivalence	P ⇔ q	if and only if “p and q are either both true or both false”

First Order Predicate Logic

Well Formed Formula (WFF)

- A sentence is called WFF if:
- It has no free variables
- All variables are properly defined by quantifiers

- Example:
- $\forall x : P(x,y) \rightarrow$ Not WFF
  Because x is defined but y is free.
- $\forall x\ \forall y: P(x) \rightarrow P(y) \rightarrow$ WFF
  Because both variables are properly defined.

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